Curriculum Vitae — Andriy V. Koval

Professional Summary

Senior Informatician and applied researcher with 10+ years of experience in data architecture, statistical modeling, and reproducible analytics across government, health services, and academia. At the Government of Alberta, serve as lead methodologist and data scientist for ministry efforts to evaluate the impact of social services, including Income Support and Career and Employment Information Services. Apply quasi-experimental research designs and reproducible research practices to connect administrative data with evidence on program outcomes.

Recent public work connects data engineering with survey research and time-series forecasting, making methods, validation, and results available for inspection. PhD in Quantitative Methods (Vanderbilt University); former tenure-track professor at the University of Central Florida. Developed reusable project infrastructure adopted across 32 repositories in GoA and delivered 20+ analyst training sessions. Author or co-author of 12+ peer-reviewed publications.

Featured Public Projects

Absent from Chronic: chronic conditions and work absenteeism. Designed to demonstrate a generic agentic analytics. Reproducible analytical workflows for a collaborative study of 63K respondents of Canadian Community Health Survey. The published analysis site connects survey-variable definitions, automated data validation, missingness and outcome diagnostics, and LASSO variable selection to the research proposal. Demonstrates traceability from raw survey files to model interpretation.

Caseload Forecast Demo: Alberta Income Support. Designed to demonstrate a simple end-to-end forecasting pipeline for testing its migration to cloud deployment. Uses 246 months of public caseload data. Methodology kept rudimentary (seasonal ARIMA with a seasonal naive baseline) to help with establishing reproducibility during migration. Versioned training-data identifiers connect forecasts to their inputs, while Parquet artifacts support portability.

Education

Dates	Degree
September 2008 – May 2014	**Doctor of Philosophy, Quantitative Methods**, Peabody College of Education and Human Development, Vanderbilt University, Nashville, TN. Dissertation: *A Graphical System for Longitudinal Modeling using Dynamic Documents: Application to NLSY97 Religiosity Data*.
August 2006 – May 2008	**Master of Arts, Quantitative Psychology**, Middle Tennessee State University, Murfreesboro, TN. Thesis: *Comparative analysis of modern differential item functioning (DIF) methods through Monte Carlo simulation*.
August 2001 – May 2005	**Bachelor of Science, Journalism**, Summa Cum Laude, with Distinction from Honors College, College of Mass Communication, Middle Tennessee State University, Murfreesboro, TN. Thesis: *Influence of Brand Advertising on the Nature of Interpersonal Communication in Ukrainian Society*.

Professional Experience

Senior Informatician

Assisted Living and Social Services (formerly Seniors, Community, and Social Services)
Strategic Data, Analytics and Business Services · Research and Open Data
June 2021 – Present

- Lead development and maintenance of the **Research Database (RDB)**, a SQL Server platform for reproducible administrative-data research
- Set team-wide standards for reproducible analysis, data quality, and documentation
- Lead methodology and analysis for social-service program evaluation using linked administrative data and quasi-experimental methods
- Develop forecasting, survey, and reporting workflows for government programs, with emphasis on validation, reproducibility, and controlled release
- Build shared analytical tools and team capability through reusable templates, staff training, and cross-ministry collaboration

Assistant Professor

University of Central Florida, College of Community Innovation and Education
Department of Health Management and Informatics
December 2018 – June 2021

- Taught graduate courses in health databases, analytics, epidemiology, and quality management; supervised researchers in applied projects
- Conducted collaborative health-services research with Veterans Affairs, state suicide-prevention partners, and community organizations

Health System Impact Fellow (CIHR)

BC Centre for Disease Control / University of British Columbia
BC Observatory for Population and Public Health
August 2017 – December 2018

- Developed and implemented decision-support methods for epidemiological reporting, coordinating implementation with regional public-health teams

Postdoctoral Research Appointment

One full-time appointment with shared supervision and overlapping affiliations:

University of Victoria, Institute of Aging & Lifelong Health
Integrative Analysis of Longitudinal Studies of Aging (IALSA) Network
August 2014 – August 2017

Vancouver Island Health Authority, Applied Clinical Research Unit
June 2015 – August 2017

University of Victoria / IALSA

- Led tools and training for reproducible, multi-site research across international teams; founded the **Data Science Studio (DSS)** for interdisciplinary learning

Vancouver Island Health Authority

- Developed an encounter-data analysis platform and advised on data governance, disclosure, and de-identification

Technical Skills

Category	Tools and Methods
Database and Architecture	SQL Server, SQLite, Parquet, relational database design, data modeling, ETL pipeline design (Ferry/Ellis patterns), data warehousing, versioned data lineage
Programming and Analysis	R (primary: tidyverse, ggplot2, Quarto, rmarkdown); SQL (proficient); Python (familiar); SAS, SPSS, Mplus, STATA
Statistical Methods	Net impact analysis, propensity score methods, quasi-experimental design, longitudinal and multilevel modeling, survey methodology, LASSO variable selection, seasonal ARIMA forecasting, holdout validation, prediction intervals
Reporting and Visualization	ggplot2, Quarto, R Markdown, interactive dashboards, Power BI, data storytelling, Edward Tufte visualization principles
Data Governance	Quality assurance standards, reproducibility protocols, data documentation (automated manifests), FOIP awareness, data disclosure methodology, de-identification
DevOps and Infrastructure	Git, GitHub, VS Code, RStudio, CI/CD workflows, AI-augmented development (Copilot, custom personas)
AI and Automation	AI-augmented research analytics, persona-based context systems, orchestration frameworks, Alberta AI Academy (Levels 1–3)

Key Projects (2021–Present)

Project	Description	Scale
Absent from Chronic	Collaborative survey research on chronic conditions and work absenteeism; validated data pipeline, predictor diagnostics, LASSO selection, and linked analytical reports	63,843 respondents; pooled CCHS cycles 2010-2011 and 2013-2014
Caseload Forecast Demo	Public Income Support forecasting sandbox; seasonal ARIMA and baseline backtesting, prediction intervals, versioned data lineage	246 monthly observations; 24-month horizon
Research Database (RDB)	Reproducible SQL Server-based analytics framework; Ferry/Ellis ETL patterns; staging and warehouse layers; automated manifests	15+ source tables, 18+ notebooks
Original FIESTA	Led study design, cohort assembly, linkage, propensity-score weighting, estimation, and reporting; findings informed funding decisions	~88,000 clients assembled; ~75,000 linked
FIESTA 2.0	Built versioned preparation pipeline and provide methodology advice; Office of Statistical Information leads linkage and estimation	734K individuals, 8.6M+ records
Quick-Start Template And Agent Harness	Adapted and extended RAnalysisSkeleton; shared project framework plus later agent-assisted engineering, modelling, and publishing	Framework used across 32 repositories
WORP Modernization	Shared survey-reporting modernization; contribute reproducible architecture, templates, and AI-assisted development	Monthly reporting workflow
Open-Data Report Preparation	Collaborated on automated IS caseload analysis, visualization, and report generation from release-approved artifacts	Preparation, not release approval
ERA Reporting	Integrated colleagues' scripts into a repeatable framework; retained their classification logic	Weekly processing; fortnightly reports

Project	Description	Scale
CEIS Impact Assessment	Standardized net impact process at three granularity levels	Methodology framework
Senior Forecasting	ASB forecasting using StatCan population projections; onboarded into template paradigm	Production system
Service System Simulation	Simulated 7-table ontology preserving structural properties of administrative data; privacy-safe external collaboration	Open-source
AIM 2025 Sandbox	AI Maximalists initiative; AI-enhanced research analytics with persona-based context system and orchestration	Capacity building
Books of Ukraine	Cross-domain data pipeline (Ferry/Ellis) for Ukrainian publishing data; multilingual, multi-contributor research collaboration	4-stage pipeline, SQLite

Selected Publications

- **Koval, A.**, Beasley, W. H., Hararuk, O., & Rodgers, J. L. (2023). Social Contagion and General Diffusion Models of Adolescent Religious Transitions: A Tutorial, and EMOSA Applications. *Journal of Research on Adolescence, 33*(1), 318–343.
- Unruh, L., Allin, S., Marchildon, G., Burke, S., Barry, S., Siersbaek, R., **Koval, A.**, & Williams, G. A. (2022). A comparison of 2020 health policy responses to the COVID-19 pandemic in Canada, Ireland, the United Kingdom and the United States of America. *Health Policy, 126*(5), 427–437.
- Andraka-Christou, B., Nguyen, T., Harris, S., Atkins, D. N., Totaram, R., Golan, O., **Koval, A.**, & Madeira, J. (2022). Harm reduction policy support among students at two US universities. *Substance Use & Misuse, 57*(8), 1185–1195.
- Kulkarni, S. A., Gurpur, V., King, C., & **Koval, A.** (2021). Impact of Gaussian Noise for Optimized Support Vector Machine Algorithm Applied to Medicare Payment on Raspberry Pi. *Informatica, 45*(4).
- Embrett, M., Liu, R. H., Aubrecht, K., **Koval, A.**, & Lai, J. (2021; published online 2020). Thinking Together, Working Apart: Leveraging a Community of Practice to Facilitate Productive and Meaningful Remote Collaboration. *International Journal of Health Policy and Management*.
- Chishtie, J. A., Babineau, J., Bielska, I. A., Cepoiu-Martin, M., Irvine, M., **Koval, A.**, … & Jaglal, S. (2019). Visual analytic tools and techniques in population health and health services research: protocol for a scoping review. *JMIR Research Protocols, 8*(10), e14019.
- Blanchette, M-A., Cheng, I., Saari, M., Ghandour, K., Aubrecht, K., Haw, J., Bailey, C., **Koval, A.**, & Sim, M. (2019). Making Contributions and Defining Success: An eDelphi Study of the Inaugural Cohort of CIHR Health System Impact Fellows. *Healthcare Policy, 15*(SP), 49–60.
- Duggan, E. C., Piccinin, A. M., Clouston, S., **Koval, A. V.**, Robitaille, A., … & Hofer, S. M. (2019). A multi-study coordinated meta-analysis of pulmonary function and cognition in aging. *The Journals of Gerontology: Series A, 74*(11), 1793–1804.
- Zammit, A. R., Piccinin, A. M., Duggan, E., **Koval, A.**, Hofer, S. M. (2019). A coordinated multi-study analysis of the longitudinal association between handgrip strength and cognitive function in older adults. *The Journals of Gerontology: Series B*.
- Moselle, K. A., Robertson, S., & **Koval, A.** (2019). "Real-World" De-Identification of High-Dimensional Transactional Health Datasets. *Studies in Health Technology and Informatics, 257*, 319–324.
- Robitaille, A., van den Hout, A., Machado, R. J., Bennett, D. A., Čukić, I., Deary, I. J., … & **Koval, A. V.** (2018). Transitions across cognitive states and death among older adults in relation to education: A multistate survival model. *Alzheimer's & Dementia, 14*(4), 462–472.
- Kelly, A., Calamia, M., **Koval, A.**, Muniz-Terrera, G., Piccinin, A., Clouston, S., … & Hofer, S. (2016). Independent and interactive impacts of hypertension and diabetes mellitus on verbal memory: A coordinated analysis. *Psychology and Aging, 31*(3), 262–273.

- **Koval, A. V.**, Beasley, W. H. (2014). Abstract: Exploring Crawford-Ferguson Family of Rotations through an Interactive Visual System. *Multivariate Behavioral Research, 49*, 288–289.
- Rodgers, J. L. & **Koval, A.** (2012). EMOSA models as developmental methodology. In Laursen, B., Little, T. & Card, N. (Eds.), *Handbook of Developmental Research Methodology* (pp. 701–724). New York: Guilford Press.

Selected Presentations (70+ total)

- **Koval, A.** (2020). Visualizing the Pandemic: Comparing Trajectories of COVID-19 across Countries. Featured speaker, VADA Program Summer School. (link)
- **Koval, A.**, Smolina, K., & Leamon, A. (2020). Using Reproducible Data Visualizations to Augment Decision-Making During Suppression of Small Counts. *International Journal of Population Data Science, 5*(5). IPDLN Conference. (link)
- **Koval, A.** (2019). Managing Data Analysis in RStudio using Project-Oriented Workflow. Invited workshop, CIHR Health System Impact Fellowship retreat, Toronto. (repo)
- **Koval, A.** (2019). Techniques for Reproducible Visualization in R: Graph as a Sequence of Three Functions. Featured speaker, VADA Program Summer School, Winnipeg. (repo)
- **Koval, A.** (2019). Using GitHub for Data Science: Version Control, Project Management, Promotion. Featured speaker, VADA Program Summer School, Winnipeg. (repo)
- **Koval, A.** (2019). Implementing Reproducible Visualizations: Application of coloring book technique in a reproducible ggplot2 system. Invited talk, UCF Department of Statistics and Data Science.
- Moselle, K. A., Robertson, S., & **Koval, A.** (2019). "Real-World" De-Identification of High-Dimensional Transactional Health Datasets. Paper presented at the 20th Annual Privacy and Security Conference, Victoria, BC.
- **Koval, A.**, Moselle, K. (2018). Clinical Context Coding Scheme: Describing Utilization of Services of Island Health between 2007–2017. **Best Data Visualization Award.** IPDLN Conference, Banff, AB. (link)
- **Koval, A.**, Moselle, K. (2018). Blueprints for Learning: How does Healthcare System advance the science of its data? CHSPR Conference, Vancouver, BC. (link)
- **Koval, A.** (2018). When notebooks are not enough. Invited talk, Matrix Institute for Applied Data Science, University of Victoria.
- **Koval, A.**, Piccinin, A., Hofer, S., Muniz, G., Beasley, W. H. (2015). A Collaborative Modeling Framework for Multi-Study Coordinated Analyses. Oral presentation, Wellcome Trust Conference, Cambridge, UK.
- **Koval, A.**, Beasley, W. H. (2017). Scientific Collaboration with GitHub. Invited speaker, Hawaii Institute of Marine Biology, Gates Coral Lab.

Collaborative Workshops and Hackathons

- *Linked Data Innovation Challenge*. **Hackathon** at the 2018 International Population Data Linkage Conference, Banff AB. Best Data Visualization award for dynamic visualization of logistic regression with the coloring book technique. (repo)
- *Probabilistic Programming for Advanced Machine Learning* **summer school** hosted by DARPA, Arlington VA, July 2017. Project demonstrated BayesDB application in simulating realistic EHR trained by transactional records. (repo)
- *Assessing the impact of different harmonization procedures*. Maelstrom Harmonization **workshop**, April 2016, Groningen, Netherlands. Role: developer of a workflow for harmonizing 5 longitudinal studies. (repo)
- *Multi-study and multivariate evaluation of healthy life expectancy*. IALSA **workshop** on multistate modeling, April 2016, Amsterdam, Netherlands. Role: modeler and repository developer. (repo)
- *Coordinated Analysis with replication using data from nine longitudinal studies*. IALSA **workshop**, February 2015, Portland, OR. Role: developing software tools for parallel analysis in Mplus, R, and GitHub. (repo)

Teaching Experience

Course	Institution	Level
Health Care Database Management (repo)	University of Central Florida	Graduate
Health Care Analytics (repo)	University of Central Florida	Graduate
Epidemiology, Analytics, and Quality Management	University of Central Florida	Graduate
Introduction to Statistical Modeling (repo)	University of Victoria	Graduate

Supervision and Mentoring

- Gabriel J. Merrin — Postdoctoral Fellow (UCF, 2019)
- Johd Pannu — Graduate Student (UCF, 2019)
- William Belanger — Graduate Student (UCF, 2020)
- Aysha Khan — Undergraduate Student (UCF, 2020)

Professional Development

- Alberta AI Academy — Levels 1, 2, and 3 (September 2025–February 2026)
- Light-touch Project Management, CIHR Health System Impact Fellowship Training, Toronto, November 2017
- Probabilistic Programming for Advancing Machine Learning, Summer School, Arlington VA, July 2017
- Advanced R Programming & R Package Development, Workshops by RStudio and Hadley Wickham, Washington DC, January 2012
- Presenting Data and Information, Workshop by Edward Tufte, San Francisco CA, May 2012
- National Longitudinal Survey Summer Workshops, Columbus OH, May 2009

Awards and Honors

- Best Data Visualization Award, International Population Data Linkage Conference, Banff AB, 2018
- Health System Impact Fellowship, Canadian Institutes of Health Research, 2017
- Travel Award, Society of Multivariate Experimental Psychology, 2013
- Outstanding Graduate Student in Quantitative Psychology, MTSU, 2008
- Summa Cum Laude, with Distinction from Honors College, MTSU, 2005

References

Available upon request.